

Effectiveness of public rural waterpoints in Bangladesh with special reference to arsenic mitigation

SUPPORTING INFORMATION

SURVEY PROCEDURES, METHODS AND MATERIALS

Sample size, location and survey procedures

Our data set combines two surveys of rural waterpoints installed by the Department of Public Health Engineering (DPHE). Initially, a Government of Bangladesh-United Nations Children's Fund (GoB-UNICEF) project, the 'Sanitation, Hygiene Education and Water in Bangladesh' (SHEWA-B), surveyed the arsenic and functional status of the 22,000 project waterpoints in 2012–13 (Figure S1). Fieldwork was conducted by 1,200 village-level Community Hygiene Promoters (CHPs) employed by the project. Following the success of this survey, DPHE, with support from UNICEF, decided to conduct a follow-up survey, the Nationwide Waterpoint Mapping Programme (NWMP) of all the waterpoints they had installed between 2007 and 2012 with support from other donors and with the government's own resources (Figure S2). Fieldwork was conducted by 1600 DPHE mechanics, based at sub-district level. To enhance the value of the database, the depth was measured at every fourth well, greatly exceeding the 1% of wells checked at SHEWA-B wells. A total of 22,000 depth measurements were made, and only these measured depths are used to evaluate depth trends.

In both surveys, lists of wells were prepared by DPHE's Management Information unit and distributed to field staff during training, which was cascaded to district level through a training-of-trainers course for the staff of the 12 DPHE zonal laboratories and UNICEF's 5 zonal WASH (water, sanitation and hygiene) officers. One DPHE chemist, the district Executive Engineer and the zonal WASH officer formed a team for day-long training events in all parts of the country. The first half of the day was given over to arsenic testing using field kits, and the

second half to field protocols and electronic (SMS) reporting procedures.

Arsenic was measured in the field by CHPs or Mechanics using the EconoQuick® test kit, which is based on the Gutzeit method and is quicker than the available alternatives. This kit has been verified for use in Bangladesh by George *et al.* (2012) and has a colour chart graduated at '0, 10, 25, 50, 100, 200, 300, 500 and 1,000' µg/L As. The colour chart has inherent limitations; analyses reported as 50 µg/L could be anywhere in the range 37.5–75 µg/L. Of 1300 SHEWA-B quality control samples analysed by atomic absorption spectrometry (AAS), 20% of samples reported as 50 µg/L exceeded this concentration. On an individual well basis, kit determinations of 'As = 50 µg/L' were either accepted at face value or checked at a laboratory. However, for calculating summary statistics, 20% of the As = 50 µg/L values are included in the exceedance percentage. Based on laboratory and field observations, it was judged that the field distinction between 0 and 10 µg/L is not robust and so results in this range were processed as a single unit '≤ 10 µg/L As'. Following national policy, the spout of each pump was painted red (>50 µg/L) or green (≤50 µg/L) based on the field test result. Note that, unless stated otherwise, statistics on arsenic contamination are the more numerous results of field kit testing.

SHEWA-B introduced a system of national waterpoint IDs, engraved on stone plates and fixed to all waterpoints. The 20-digit ID (see later) contains the year of installation, geocoding to village level, land type, agency, waterpoint type, and serial number. During the NWMP, IDs pre-printed on plastic tags were attached to each well at the time of testing. This publicly visible tag promotes transparency and accountability, and facilitates monitoring.

Reporting was made by SMS to DPHE's database server, and under the NWMP a small payment of Taka 30 (\$0.4) was paid directly to the mechanics using the

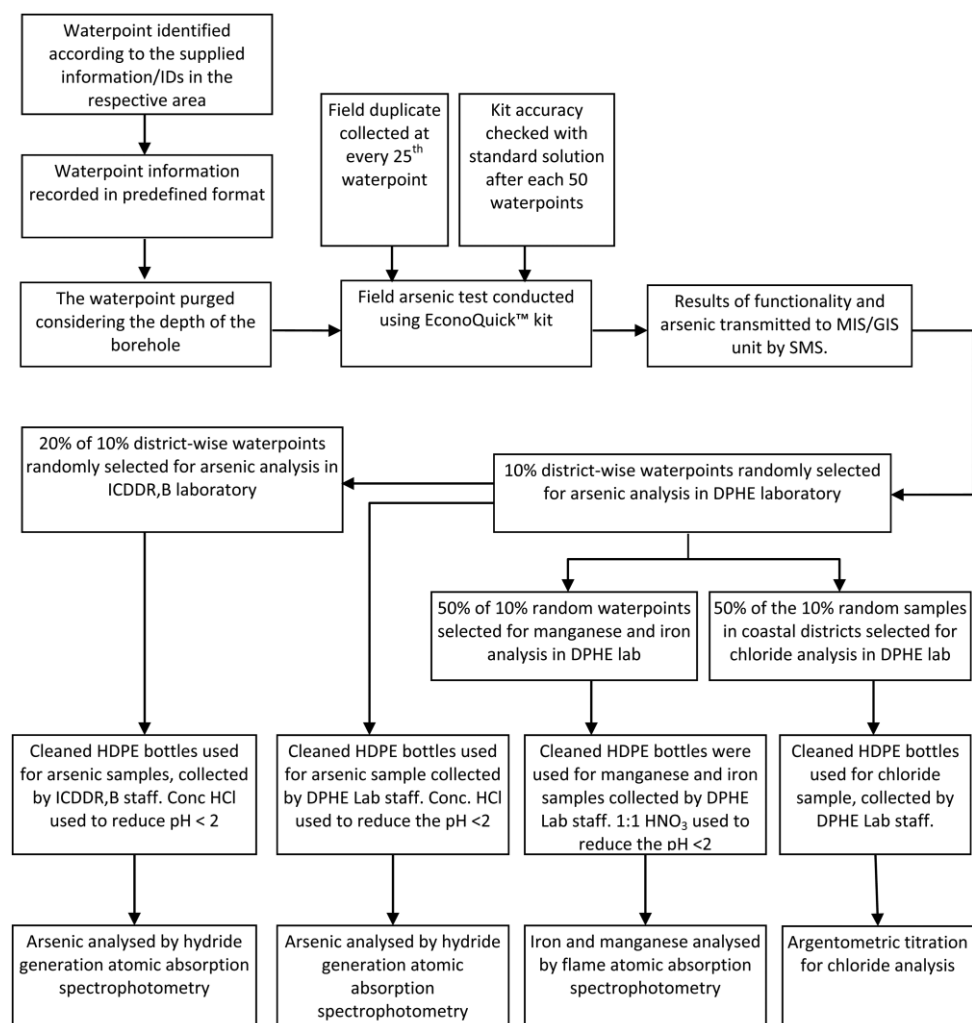


Figure S1 | Flow diagram for surveys of SHEWA-B waterpoints.

bKash™ (a joint venture of BRAC Bank Limited and Money in Motion LLC) mobile banking system for each valid SMS report received, which can be encashed through 10,000 agents across the country.

While the SHEWA-B survey recorded waterpoints as simply functional or non-functional, the NWMP further differentiated between temporarily or permanently non-functional, i.e. according to whether they were expected to be fixed or were effectively abandoned, although not necessarily irreparably.

Year round availability of water was assessed by combining the distribution of suction mode pumps with DPHE's dry season water level monitoring, in which the

depth to water (below top of casing) is measured three times in one operational shallow tubewell and, if present, one deep tubewell in each union during March and April. The deepest value is then reported and compiled by headquarters. A threshold of 25 feet (7.6 m) was adopted for the onset of serious difficulty or failure in drawing water. Estimates were derived for SHEWA-B project wells in 2012, based on 1,838 DTW6s, 2,364 STW6s and 1,586 RW6s (see Table S1 for definitions) in around 450 unions, covering all parts of the country. The proportions calculated were applied to the national data set (Table S2). However, it should be noted that functionality was recorded at all times of year, including March and April, although there may be

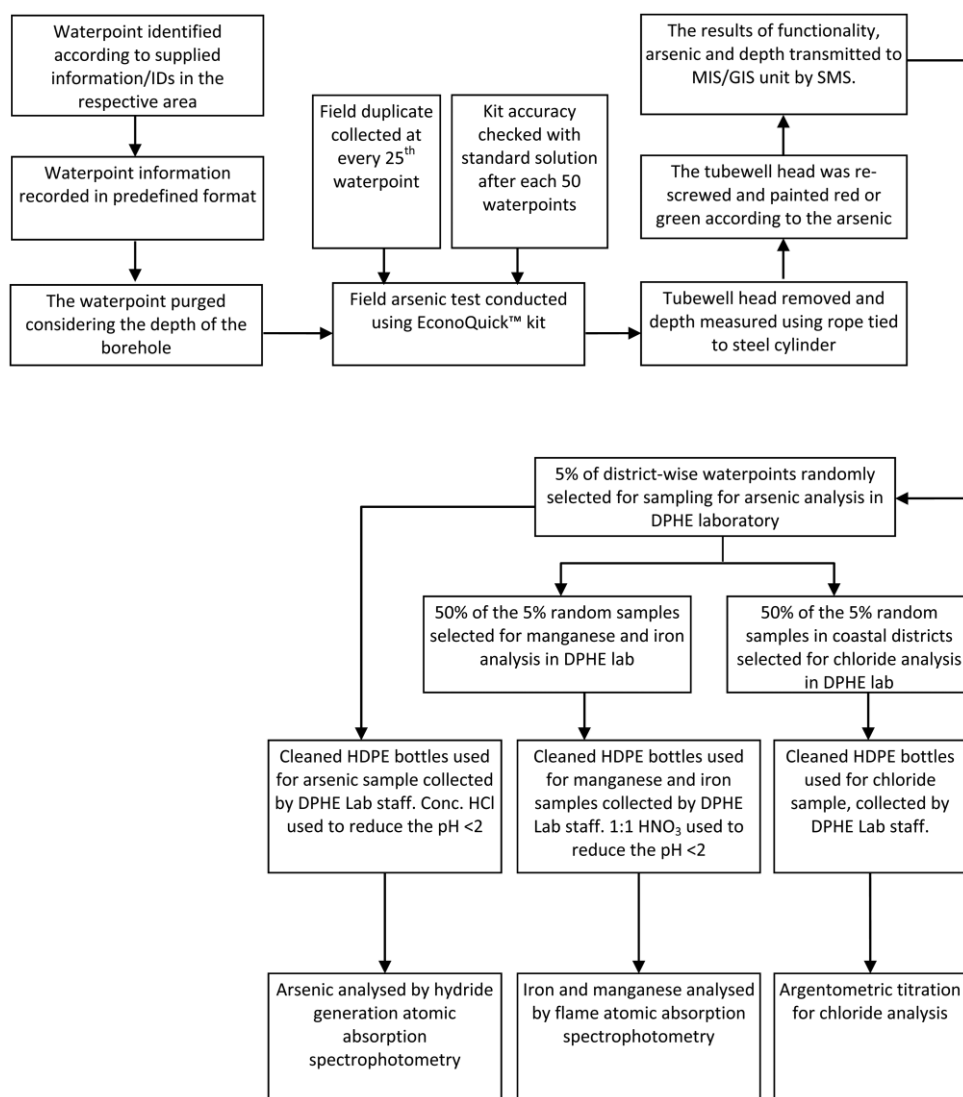


Figure S2 | Flow diagram for nationwide waterpoint mapping programme.

some double counting with suction-mode wells classed as ‘temporarily non-functional’ and measured in March or April.

The ‘adequate quantity’ and ‘large number’ of people were captured in the numbers of households using the device. The non-functionality rate, both overall and averaged for years 5 to 7, is adopted as an indicator of the reliability and sustainability of the technology. The criteria ‘safe and good’ are measured through water quality parameters and the question as to whether people actually drink the water. Information on construction costs was obtained from project documents.

Quality control and additional water quality analyses

Previous water quality surveys have been plagued by inadequate quality assurance. To ensure confidence, both surveys included two tiers of quality control (QC), which was explained to field surveyors during training. At each stage, QC sites were selected only after sampling and testing had been completed. In the SHEWA-B project, 10% of waterpoints were randomly selected for sampling by the DPHE zonal laboratories (As 1,187; Fe & Mn 577; Cl 438); and 20% of the 10% subset was randomly selected for independent sampling and arsenic analysis by the International

Table S1 | Characteristics of rural waterpoints

Acronym	Name	Description	Applications	Typical cost ² (\$)	Notes
DTW6	Deep tubewell with Nr 6 (suction) pump	Manually drilled well >150 m and <350 m deep with 38 mm diameter casing and typically c. 6 m of slotted casing.	Avoidance of arsenic or salinity in shallow aquifers	\$800–900 ² (275 m)	Nominal DTWs often <150 m in practice.
DTW-T	Deep tubewell with Tara or Tara-Dev pump	As above but with 20–25 m of 75 mm casing in upper part to accommodate a pump piston.	As above in areas where water levels fall below about 7.5 m.	\$900–1000 ² (275 m)	
STW6	Shallow tubewell with Nr 6 (suction) pump	Manually drilled well <150 m deep, usually much less, with 38 mm diameter PVC casing, and typically 3–6 m of slotted casing. Very cheap, and easy to install, operate and maintain.	Most widely used option, but constrained by arsenic and declining water levels.	\$100–150 ² (60 m)	
STW-T	Shallow tubewell with Tara or Tara-Dev pump	As above but with 20–25 m of 75 mm in upper part to accommodate pump piston.	Replaces STW6 where water level >7.5 m and not contaminated.	\$300–350 ² (60 m)	
RW6	Ringwell with Nr 6 (suction) pump	Reinforced concrete rings, about 1 m in diameter, sunk to 5–15 m. A wall and mesh provide surface protection. Abstraction via an offset hand-pump.	Traditional technology sometimes advocated for avoiding arsenic.	\$650–700 ² (13 m with RCC slab)	
RW-T	Ringwell with Tara or Tara-Dev pump	Subsurface construction as above, however, the well is capped by a reinforced concrete (RCC) slab to support the weight of the Tara-Dev assembly directly above the slab.	As above but where water levels fall below 7.5 m.	\$750–800 ² (13 m with RCC slab)	
SST/ VSST	(Very) shallow shrouded tubewell	Similar to the STW-Nr 6, but shallower and drilled at larger diameter to allow placement of a sand pack (shroud) against fine sand aquifers.	Principally used to tap shallow fresh water lenses, and occasionally advocated for avoiding arsenic.	\$100–125 ²	
RWH	Rain water harvester	3,000 litre masonry tank collects water from a single roof to supply drinking water for a single family.	Mostly installed in areas where fresh groundwater is unavailable or polluted.	\$300	Sizing based on 10–20 L/c/d of water only.
PSF	Pond sand filter	A small slow sand filter and clear water reservoir. Input water is drawn from a pond using a Nr 6 pump.	The PSF works best when is combined with a roughing filter.	\$500 ²	Requires significant O&M effort
Inf. Gal	Infiltration gallery	A trench or sump abstraction constructed adjacent to a stream, equipped with a Nr 6 pump.	A rarely used technology, applied where others unsuccessful.	\$250	

(continued)

Table S1 | continued

Acronym	Name	Description	Applications	Typical cost ² (\$)	Notes
GFS	Gravity flow system	Small scale piped water system drawing water from streams or springs at distances of the order of a kilometre, and treated by slow sand filter close to point of use, with water collected from 10–20 standpipes.	Used in Chittagong Hill Tracts to supply communities where bedrock is of very low permeability and/or inaccessible to drilling rigs.	\$25,000–30,000	

Notes:

1. Unlike the Nr 6 pump, which operates under suction, the Tara or Tara-Dev is a force mode pump. The original Tara (Bengali for 'star') had a vertically acting plunger mechanism; the Tara-Dev, which has largely replaced it, has a horizontal T-bar, and is similar to the AfriDev and India Mark II hand-pumps.

2. Based on Taka costs from the Revised Development Project Proforma for SHEWA-B converted at \$1 = Tk80. Costs include apron or platform where applicable.

3. All groundwater extraction devices are installed by manual drilling / excavation techniques.

4. Further details of these technologies are given by Ahmed (2002).

Table S2 | Number of devices installed by year

WP Type	2006	2007	2008	2009	2010	2011	2012	Total
DTW-6	1,149	4,122	10,553	5,457	5,461	8,213	4,111	39,066
DTW-T	387	410	570	671	1,272	3,329	1,121	7,760
STW6	347	2,231	957	4,296	8,340	4,981	2,996	24,148
STW-T	287	1,749	1,756	2,197	2,595	6,033	2,669	17,286
RW6	177	803	1,123	852	1,239	1,086	938	6,218
RW-T	10	176	352	186	514	376	94	1,708
SST/VSST	115	1,136	595	293	517	446	276	3,378
RWH	24	111	18	12	64	574	211	1,014
PSF	3	149	117	399	588	623	198	2,104

Note. These data relate only to waterpoints in the NWMP survey.

Centre for Diarrhoeal Disease Research – Bangladesh (ICDDR,B). Under NWMP, 5% of wells (3,110) were sampled for arsenic analysis at the DPHE laboratories, and 10% of these sites selected for independent QC checks and analysis of arsenic at an external laboratory. At approximately 50% of the QC sites, samples were collected for iron and manganese (1,565), and in southern Bangladesh also chloride (631), analysis at the respective DPHE zonal laboratories.

Before sampling, tubewells were purged according to their depth. Standard high density polythene (HDPE) sample bottles were used for sample collection. Sample bottles were cleaned in the laboratory, initially with tap water, then with 1:1 nitric acid followed by distilled water. After drying, the sample bottles were then labelled. Concentrated hydrochloric acid was used to preserve samples for arsenic

analysis, and 1:1 concentrated nitric acid was added to preserve samples for iron and manganese analysis. Acids were analytical grade chemicals from Merck, Aldrich and BDH. Sufficient acid was added to reduce the pH to below 2. Arsenic was determined by hydride vapour generation atomic absorption spectrometry (AAS) following the standard operating protocol of the DPHE laboratory for arsenic analysis. Manganese and iron were analysed by flame atomic absorption spectrometry. Two models of AAS machine were used in different zonal laboratories: Simadzu and Varian. The method detection limit (MDL) for arsenic analysis by Shimadzu model A7000 (HVG) was 0.0007 mg/L, and the MDL by Varian model AA220 (HVG) was 0.0005 mg/L. Chloride was determined by argentometric titrations.

Most samples were analysed within one month of collection. ICDDR,B analysed arsenic by Atomic Absorption Spectrometry Model AA-7000 (HVG-1), MDL 0.0005 mg/L. Note that the QC process does not permit precise comparison of the field and laboratory analyses or of the two laboratory analyses because the tests were done on different samples. Hence any comparisons evaluate the total sampling and analytical process and assume no real change in water quality between sampling events.

Sanitary inspections were conducted by DPHE Chemists during chemical QC sampling and were an extended form of WHO's (1997) standard 10 point questionnaire that included questions and discussion with the users about previous arsenic

testing, marking, and water use such as the number of families using the device and whether the water was actually used. Geographical coordinates were measured by Global Positioning System (GPS) during the survey. A supplementary survey measuring *E. coli* in the field using membrane filtration and compact dry plate (Nissui EC 240) culture at 1035 SHEWA-B QC sites as used in the 2012–13 Multiple Indicator Cluster Survey (MICS) (Kodaka *et al.* 2006; Bangladesh Bureau of Statistics [BBS]/UNICEF 2014). The procedure, and the one-day training course given to DPHE zonal chemists, was adapted from that used in the 2012–13 MICS (BBS/UNICEF 2014). Before sampling for microbiological testing, pump spouts or taps were sterilised by flaming with alcohol. Plate

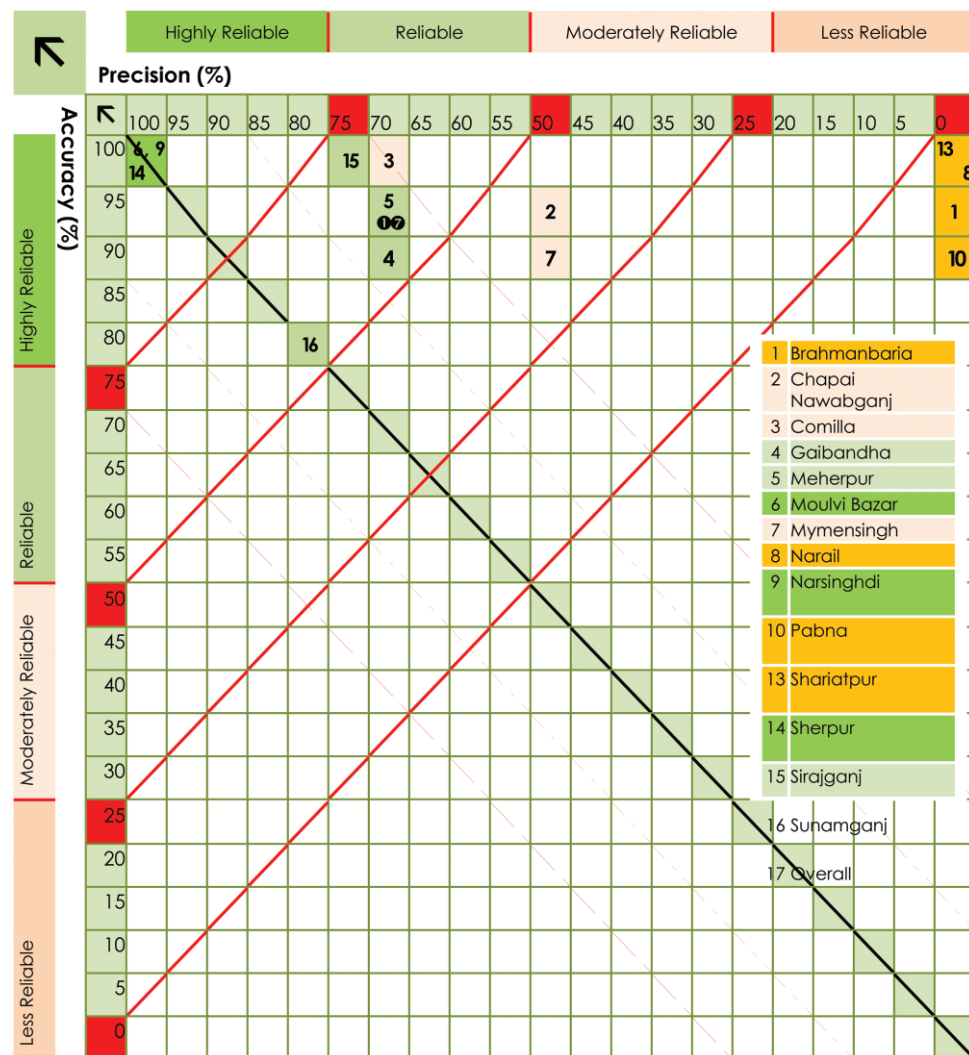


Figure S3 | District-wise reliability of arsenic field kit testing.

counts were made on 24-hour cultures from 1 and 100 ml water samples. If the result from the 100 ml sample was ≤ 100 colonies this value was accepted, but if > 100 colonies the result from the 1 ml sample was adopted.

QUALITY CONTROL RESULTS

Comparison of field test kit and DPHE laboratory analyses for arsenic

District-wise precision and accuracy were calculated by using the following equation:

Matrix for calculation of precision and accuracy			Condition determined by laboratory		
			True	False	
			LAB Indicates water is contaminated (As > 50 $\mu\text{g/L}$)	LAB Indicates water is NOT contaminated (As ≤ 50 $\mu\text{g/L}$)	
Test outcome	Positive	Kit Indicates water is contaminated	True positive	False positive	Positive predictive value or Precision
	Negative	Kit Indicates water is NOT contaminated	False negative	True negative	Negative predictive value Accuracy
			Sensitivity or recall	Specificity (or its complement, Fall-Out)	Accuracy

The results are presented in [Figure S3](#), and are divided into four categories: highly reliable, reliable, less reliable and unreliable. In calculating precision and accuracy, laboratory testing results were taken to be correct. It is unexpected that laboratory and field test results will be the same because of the limitations of different method of analysis, instrumentation, human and instrumental error, level of detection, etc. Data from some districts or sub-districts classified as unreliable, such as parts of Narail, Shariatpur and Brahmanbaria, have been scheduled for retesting. This is anticipated to increase the number of exceedances but not to have a major effect on the overall statistics.

The results indicated that overall the initial round of field arsenic testing had an accuracy of 79% and a precision of 76%. These results, together with other information were used to determine where retesting was required.

[Figure S4](#) compares the results of analyses performed in the DPHE and ICDDR,B laboratories.

$$\text{Accuracy} = \left\{ \frac{\text{Number of true positives} + \text{Number of true negatives}}{\text{Number of true positives} + \text{False positives} + \text{False negatives} + \text{Number of true negatives}} \right\} \times 100$$

$$\text{Precision} = \left\{ \frac{\text{Number of true positives}}{\text{Number of true positives} + \text{False positives}} \right\} \times 100$$

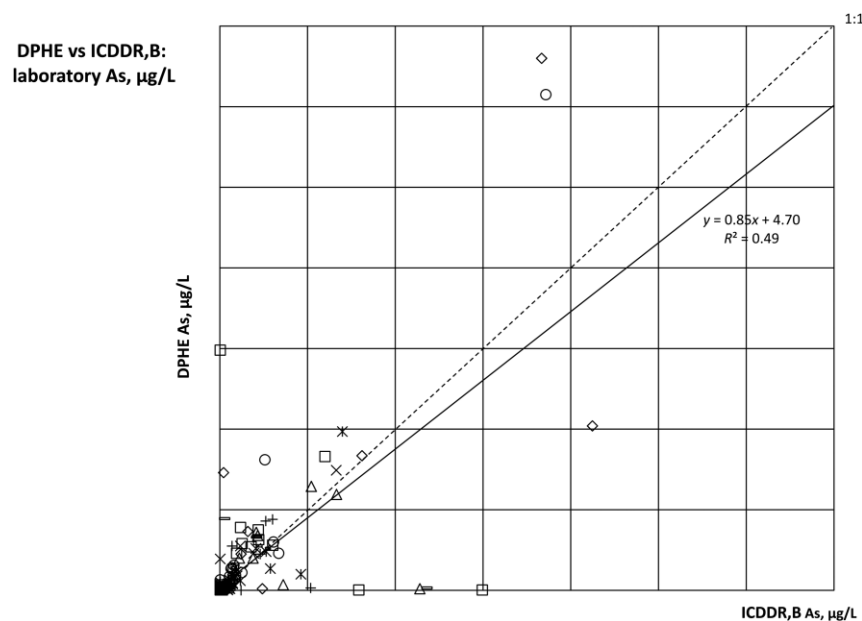


Figure S4 | Comparison of DPHE and ICDDR,B laboratory analyses for arsenic.

ELECTRONIC DATA TABLES [NWMP_AND_SHEWAB SURVEY DATA.XLSX]

Tab	Content
1. WP data	List of waterpoint IDs and field recordings of arsenic content, functionality, and depth. Note the ID contains information on location, waterpoint type, age, land use, and funding agency.
2. ID coding	Explanation of the meaning of the ID code
3. Geocoded locations	List of unique district, upazila and union names in the database linked to the geocode contained in the WP ID.
4. Chemical QC & sanitary insp.	Determinations of As, Fe, Mn & Cl in DPHE laboratories
5. Sanitary codes	Explanation of sanitary inspection codes
6. Microbiology	Colony counts for <i>E. coli</i> and Other Coliforms in 1 ml and 100 ml sample plates at SHEWA-B waterpoints
7. Supplementary questions	Supplementary questions asked during chemical QC sampling of SHEWA-B waterpoints including the painting of the pump spout to reflect arsenic testing; the number of families using the device; and whether users actually drink the water.
8. ICDDR,B arsenic	Determinations of As in ICDDR,B laboratory for the Tier 2 QC of SHEWA-B waterpoints.

REFERENCES

- Ahmed, M. F. 2002 Alternative water supply options in Bangladesh. Theme paper. In International Workshop on Arsenic Mitigation in Bangladesh, Dhaka. Ministry of Local Government, Rural Development and Cooperatives, 86p.
- BBS/UNICEF 2014 (in press) Multiple Indicator Cluster Survey (MICS) 2012-13. Bangladesh Bureau of Statistics and UNICEF, Bangladesh.
- George, C. M., Zheng, Y., Graziano, J. H., Rasul, S. B., Mey, J. L. & van Geen, A. 2012 [Evaluation of an arsenic test kit for rapid well screening in Bangladesh](#). *Environ. Sci. Technol.* **46**, 11213–11219.
- Kodaka, H., Mizuochi, S., Teramura, H. & Nirazuka, T. 2006 Comparison of the compact dry EC with the most probably number method (AOAC Official Method 966.24) for enumeration of *Escherichia coli* and coliform bacteria in raw meats. *J. AOAC Int.* **89**, 100–114.
- WHO 1997 *Surveillance and Control of Community Supplies*. World Health Organization, Geneva.